

- 8 What length of wire of resistivity $2\pi \times 10^{-4} \Omega \text{ m}$ and having a diameter of 1.0 mm is required to make a 20.0Ω resistor?
- 9 A metal rod has a resistivity of $\pi \times 10^{-3} \Omega \text{ m}$. What diameter rod should be used so that a 0.5 m length has a resistance of 5.0Ω ?
- 10 A wire has a resistance of 100Ω . If the wire is drawn out to twice its original length, and its diameter is halved, what is its new resistance?
- 11 A nichrome wire has a resistivity of $1.0 \times 10^{-6} \Omega \text{ m}$ and a diameter of 0.40 mm. What length of wire is needed to carry a current of 30 mA when there is a potential difference of 12 V across it?
- 12 A wire is 1.0 m long and has a resistance of 1.05Ω . A wire 50.0 m long is made from the same material, and has twice the diameter. Calculate the resistance of the wire.

Answers

1 $8.0 \times 10^1 \text{ V}$	2 $5 \times 10^5 \text{ V}$	3 0.2Ω
4 $5.0 \times 10^{-3} \text{ A}$	5 2 V	6 12 V
7 $5 \times 10^{-3} \Omega \text{ m}$	8 $2.5 \times 10^{-2} \text{ m}$	9 $2 \times 10^{-2} \text{ m}$
10 800Ω	11 $5 \times 10^1 \text{ m}$	12 13Ω

Series and parallel resistances

An electric circuit is made up of resistors in series, parallel or a combination of these.

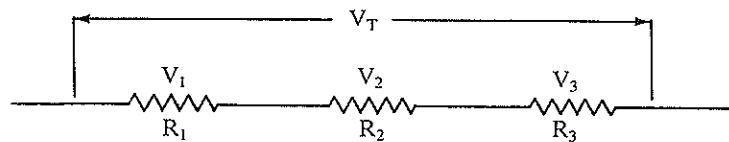
In a series circuit any charge that flows through one conductor must flow through all conductors; hence, the current flowing through each resistor is the same. The total resistance for the circuit is

$$R_T = R_1 + R_2 + R_3 + \dots$$

where R_T is the total resistance of the circuit and $R_1, R_2, R_3 \dots$ are the individual resistances.

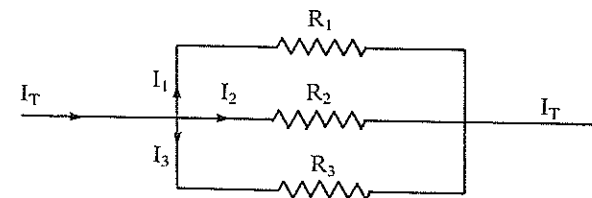
The total potential difference applied across the resistances divides proportionally among the individual resistors so that the sum of each potential drop across each resistor equals the total potential drop across the circuit.

That is



$$V_T = V_1 + V_2 + V_3$$

In a parallel circuit the current divides where the circuit branches. The current flowing through each branch depends on the branch resistance. If I_T is the total current in a circuit then



$$I_T = I_1 + I_2 + I_3 \dots$$

where I_1, I_2, I_3 are the individual currents within the network.

All resistors within a parallel branch have the same potential difference across each of the resistors.

For any branch if $I_T = I_1 + I_2 + I_3 + \dots$

$$\text{and } I_1 = \frac{V}{R_1}, \quad I_2 = \frac{V}{R_2}, \quad I_3 = \frac{V}{R_3}, \dots$$

$$\text{then } \frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

where R_T is the equivalent resistance of the parallel branch of resistors $R_1, R_2, R_3, \dots$

Example 4

A 5.0Ω resistor and a 15.0Ω resistor are connected in series and a $1.0 \times 10^2 \text{ V}$ potential difference is applied across them. Calculate:

- the total resistance of the circuit
- the current flowing through each resistor
- the potential drop across each resistor.

$$R_1 = 5.0 \Omega$$

$$R_2 = 15.0 \Omega$$

$$V = 1.0 \times 10^2 \text{ V}$$

$$R_T = ?$$

$$I = ?$$

$$V_1 = ?$$

$$V_2 = ?$$

$$\begin{aligned} \text{(a) } R_T &= R_1 + R_2 \\ &= 5.0 + 15.0 \\ &= 20.0 \Omega \end{aligned}$$

$$\begin{aligned} \text{(b) } I &= \frac{V}{R} \\ &= \frac{1.0 \times 10^2}{20.0} \\ &= 5.0 \text{ A} \end{aligned}$$

$$\begin{aligned} \text{(c) Drop across } R_1 \text{ is} \\ V_1 &= IR_1 \\ &= 5.0 \times 5.0 \\ &= 25 \text{ V} \end{aligned}$$

$$\begin{aligned} \text{Drop across } R_2 \text{ is} \\ V_2 &= IR_2 \\ &= 5.0 \times 15.0 \\ &= 75 \text{ V} \end{aligned}$$

Check the total voltage = 100 V

$$\begin{aligned} V_T &= V_1 + V_2 \\ &= 25 + 75 \\ &= 100 \text{ V} \end{aligned}$$

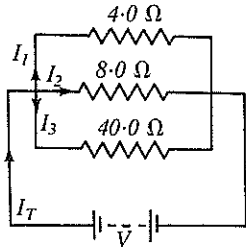
The total resistance is 20.0 Ω , the current flow is 5.0 A and the potential drop across each resistor is 25 V and 75 V.

Example 5

Three resistors of 4.0 Ω , 8.0 Ω and 40.0 Ω are connected in parallel. A potential difference of 8 V is applied across the branch. Calculate:

- the equivalent resistance
- the current in each resistor
- the total current entering the branch.

$$\begin{aligned} R_1 &= 4.0 \text{ } \Omega \\ R_2 &= 8.0 \text{ } \Omega \\ R_3 &= 40.0 \text{ } \Omega \\ V_T &= 8 \text{ V} \\ R_T &= ? \\ I_1 &= ? \\ I_2 &= ? \\ I_3 &= ? \\ I_T &= ? \end{aligned}$$



$$\begin{aligned} (a) \quad \frac{1}{R_T} &= \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \\ &= \frac{1}{4.0} + \frac{1}{8.0} + \frac{1}{40.0} \\ &= \frac{10 + 5 + 1}{40.0} \\ &= \frac{16}{40.0} \\ R_T &= 2.5 \text{ } \Omega \end{aligned}$$

(b) If $V = 8 \text{ V}$

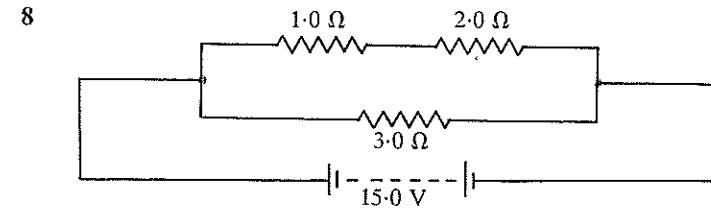
$$\begin{aligned} \text{then } I_1 &= \frac{V}{R_1} \\ &= \frac{8}{4.0} \\ &= 2 \text{ A} \\ I_2 &= \frac{V}{R_2} \\ &= \frac{8}{8.0} \\ &= 1 \text{ A} \\ I_3 &= \frac{V}{R_3} \\ &= \frac{8}{40.0} \\ &= 0.2 \text{ A} \end{aligned}$$

$$\begin{aligned} (c) \quad I_T &= I_1 + I_2 + I_3 \\ &= 2 + 1 + 0.2 \\ &= 3.2 \text{ A} \\ I_T &= 3 \text{ A} \end{aligned}$$

The equivalent resistance is 2.5 Ω , the current in each resistor 2 A, 1 A and 0.2 A respectively, and the total current in the branch 3 A.

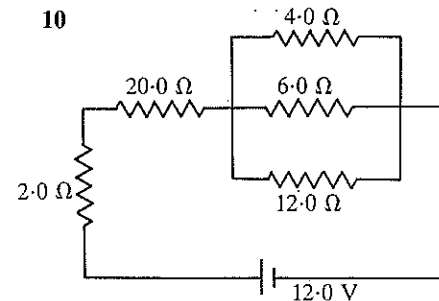
Set 36

- What single resistor could replace 20 Ω , 30 Ω and 50 Ω resistors which have been joined in series?
- What resistor placed in series with a 200 Ω resistor will produce a 1.0 k Ω resistor?
- What single resistor could replace three 30 Ω resistors joined in parallel?
- Find the equivalent resistance when 12 Ω , 6 Ω and 4 Ω are arranged
 - in series, and
 - in parallel.
- What resistance must be placed in parallel with an 11 Ω resistor to reduce its resistance to $1.0 \times 10^1 \text{ } \Omega$?
- Three resistors, having resistances of 20, 30 and 50 Ω , are connected in series across a 200 V supply. Find:
 - the total resistance
 - the current flowing in each resistor
 - the potential difference across each resistor.
- A lamp operates on a 100.0 V supply and draws a current of 0.50 A. What resistor must be placed in series if it is to be used on a 230 V supply?



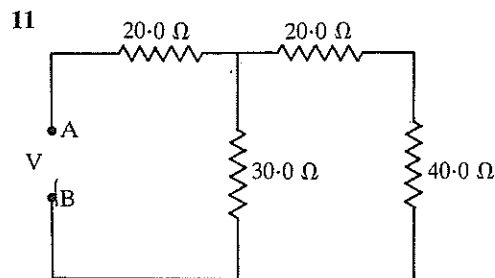
Calculate the current in the 2 Ω resistor.

- Three resistors of 4.0 Ω , 8.0 Ω and 40.0 Ω are connected in parallel. The 4.0 Ω resistor carries a current of 2 A. Calculate:
 - the combined resistance
 - the potential difference across the parallel branch
 - the current in the 8.0 Ω and 40.0 Ω resistors.



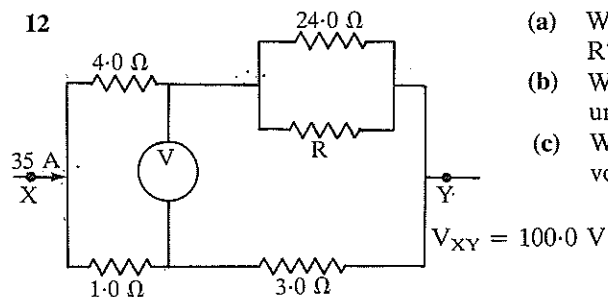
The diagram shows resistances joined in parallel and in series.

- What is the equivalent resistance of the circuit?
- What current flows through the 20 Ω resistor?
- What is the potential difference across the 2 Ω resistor?
- What is the potential difference across the 4 Ω resistor?
- What is the current through the 4 Ω resistor?

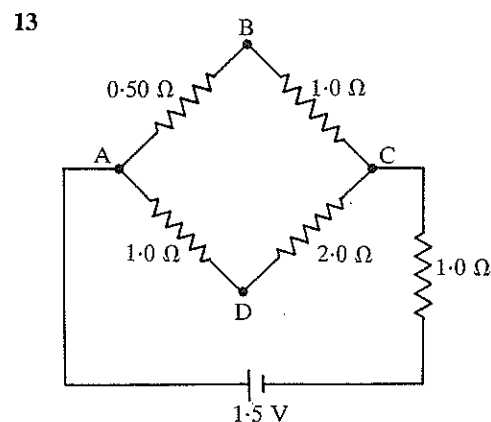


If there is a current of 1.0×10^2 mA in the 40Ω resistor, calculate:

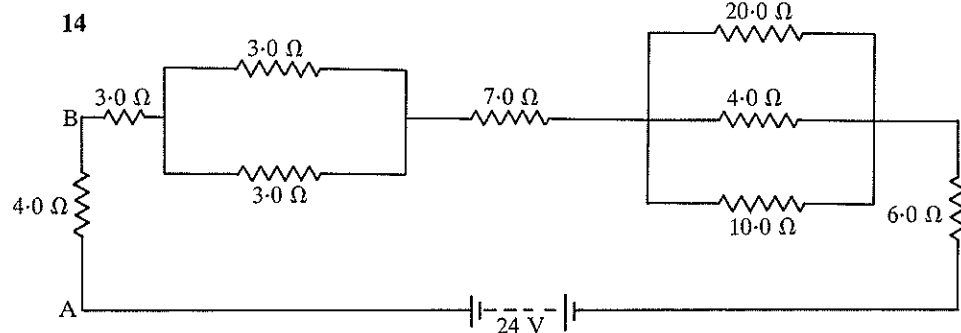
- V_{AB}
- I_{AB}
- the current in the 30Ω resistor.



- What current flows through R?
- What is the value of the unknown resistance R?
- What is the reading on the voltmeter V?



- Find the resistance of the network between A and C.
- Find the currents I_{ABC} and I_{ADC} .
- Find V_{AB} and V_{AD} .
- Find potential difference between B and D.
- If a galvanometer were connected between B and D, would it register a current?



Calculate:

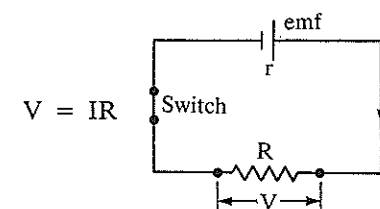
- the current from the battery
- the potential difference between A and B.

Answers

- | | | | | | |
|----|-----------|---|--------|-------|-------------------|
| 1 | 100 Ω | 2 | 800 Ω | 3 | 10 Ω |
| 4 | (a) 22 Ω | (b) | 2 Ω | 5 | 110 Ω |
| 6 | (a) 100 Ω | (b) | 2 A | (c) | 40 V, 60 V, 100 V |
| 7 | 260 Ω | 8 | 5 A | | |
| 9 | (a) 2.5 Ω | (b) | 8 V | (c) | 1 A, 0.2 A |
| 10 | (a) 24 Ω | (b) | 0.5 A | (c) | 1 V |
| | (d) | 1 V | (e) | 0.2 A | |
| 11 | (a) 12 V | (b) | 0.30 A | (c) | 0.20 A |
| 12 | (a) 7.5 A | (b) | 8 Ω | (c) | 15 V |
| 13 | (a) | $R_{ABC} = 1.5 \Omega, R_{ADC} = 3.0 \Omega, R_{AC} = 1.0 \Omega$ | | | |
| | (b) | $I_{ABC} = 0.50 \text{ A}, I_{ADC} = 0.25 \text{ A}$ | | | |
| | (c) | $V_{AB} = 0.25 \text{ V}, V_{AD} = 0.25 \text{ V}$ | | | |
| | (d) | 0.00 V | | | |
| | (e) | No | | | |
| 14 | (a) | 1.0 A | (b) | 4.0 V | |

Potential and potential difference

The electromotive force (emf) has the value of the potential difference between the terminals of a battery or generator when it is not supplying current. The potential difference across the external resistance in the circuit illustrated is given by



If r is the internal resistance of the cell of electromotive force then

$$\text{emf} = I(R + r)$$

$$= V + Ir$$

$$\therefore V = \text{emf} - Ir$$

Since Ir is the voltage drop across the internal resistance of the cell then

$$\text{Terminal voltage} = \text{emf} - \text{Voltage drop in internal resistance}$$